

The next processor that holds promise for tablets is multiprocessor

The STAMP journey began in 2009 as a quest to create a low-cost multimedia device for applications like Internet radio, Internet photo frames and home energy management consoles. They developed their previous IP—a reference design for Internet radio and digital photo frames—to make a low-cost tablet.

Darshak Vasavada, CEO, STAMP Computers, spoke to Abhishek Mutha of EFY about technical challenges while designing a tablet, resistive versus capacitive screen, the latest trends and applications for tablets



DARSHAK VASAVADA
CEO, STAMP COMPUTERS

How do you see the role of tablets evolving, especially in India?

People are figuring out more and more new things to do with the tablet, and it is finding use almost everywhere. If we look at the rural market where PCs couldn't go due to power issues and accessibility problems, cell phones and other mobile devices have penetrated easily. Now even tablets are making their way into rural India. In fact, tablets are available at price points similar to phones. Every day we are seeing more and more creative uses for tablets, and I am sure the market is going to be very exciting.

What are the technical trends in tablets?

Most tablets today use ARM Cortex-A8 processor. The next processor that holds promise for tablets is Cortex-A9. A9 is a multi-core processor, which allows a dual-core or quad-core functionality while running at 1 or 1.5 GHz.

There is another new trend called 'big little architecture.' In this architecture, there is one big processor and one small processor for optimising the battery usage. When the load on the processor is low, the small processor kicks in, allowing the device and memory to be kept alive while not consuming too much power.

What to expect on the application front?

Consider a tablet for field work. It gives an extension to enterprise resource planning (ERP) system of the company. Today, ERP system is limited to computers within the company. Use of tablets allows companies to stretch their ERP system, so their sales force can convert deals right at the client's office without having to come back to the office to finalise it. For a device like this, a very long battery life and a variety of connectivity options are required so that the user is not bogged down by charging or connection issues.

It also has to support a lot of peripherals so that connection to external devices is easy and as required.

What are the specifications of STAMP tablet?

We designed a rugged casing since this device is targeted at field applications. We have also incorporated a full USB port, which is a necessity for connectivity with other devices. It provides full 50 mA of current, so multiple devices can be connected to it simultaneously. A cellphone can also be charged through this USB port.

The tablet uses an ARM Cortex-A8 processor clocked at 800 MHz, and we plan to over-clock it to 1 GHz pretty soon. It has a capacitive multi-touch screen with a 24-bit RGB display, which provides full-size video at WVGA resolution. Moreover, it has a provision for SD card and power and headphone connectivity options. It is a pretty simple tablet that can play five hours of video on a single charge.

What technical challenges did your design team face while working on this device?

We came across a problem whose solution finding process was pretty interesting. In our motherboard, there are two signals that are exchanged between the processor and the memory—row-address strobe (RAS) and column-address strobe (CAS).

SDRAM is notationally organised into a grid-like pattern with rows and columns. Data stored in the SDRAM comes in blocks, defined by coordinates of the row and column of the specific information.

During design, RAS and CAS were flipped, which brought up some problems. When we got the board from the factory for the first time, we had no idea whether the board was fully functional or not. The first thing to do was to bring up the memory. Due to the issue with the flipped